

Software Engineering Internship Assignment

Build a production-grade Bike Auction Platform where registered users can participate in live auctions for used motorcycles.

The platform should support multiple simultaneous auctions, real-time bidding, auction lifecycle management, and operational observability expected in a modern production system.

The application should provide a modern, intuitive, and user-friendly experience for both buyers and administrators.

You are free to make reasonable assumptions where requirements are not explicitly specified.

Expectations

Your submission should demonstrate:

- Strong engineering fundamentals
- Clean and maintainable code
- Thoughtful system design
- Production-grade architecture
- Scalability considerations
- Security best practices
- Logging, monitoring, and observability
- Automated testing
- Good API design
- Excellent user experience and interface design

Deliverables

- Source code
- Setup instructions
- Architecture/design document
- Deployment instructions
- Assumptions and trade-offs made during implementation

We are more interested in the quality of engineering decisions and execution than the number of features implemented.